

Reaction kinetics

A Level Chemistry

Rate equations and orders

A **rate equation** 速率方程 shows how the rate depends on the concentrations of the reactants:

$$\text{rate} = k [\text{A}]^m [\text{B}]^n$$

- m is the **order of reaction** 反应级数 with respect to A, and n the order with respect to B. Each is 0, 1 or 2.
- the **overall order of reaction** 总反应级数 is $m + n$.
- k is the **rate constant** 速率常数. The rate equation can only be found by experiment, not from the balanced equation.

Finding the order

- **initial rates method**: change one concentration at a time and see how the starting rate changes. If doubling [A] doubles the rate, the order in A is 1; if it quadruples the rate, the order is 2; if the rate is unchanged, the order is 0.
- **graphs**: a concentration–time graph for a first-order reaction has a constant **half-life** 半衰期 (the time for the concentration to halve). A rate–concentration graph is a straight line through the origin for first order, and a curve for second order.

Half-life and the rate constant

For a **first-order** reaction the half-life is **constant** —it does not depend on the concentration. You can find the rate constant from it:

$$k = \frac{0.693}{t_{\frac{1}{2}}}$$

You can also find k by putting initial-rate data into the rate equation.

Reaction mechanisms

Most reactions happen in several steps. The slowest step is the **rate-determining step** 决速步骤, and it controls the overall rate. Only the species involved up to and including this step appear in the rate equation.

- an **intermediate** 中间体 is a species made in one step and then used up in a later step. It is not in the overall equation.
- you can suggest a mechanism that fits both the rate equation and the overall equation, predict the order from a given mechanism, or pick out the rate-determining step.

If you compare the **initial rate** 初始速率 of different mixtures, you can deduce the rate equation, and from that work out the mechanism.

Effect of temperature

Raising the temperature increases the rate constant k (more molecules pass the activation energy), so the rate goes up.

Catalysts

Catalysts 催化剂 can be homogeneous or heterogeneous.

Heterogeneous catalysts

A **heterogeneous catalyst** 多相催化剂 is in a different physical state from the reactants (usually a solid with gases). It works in three stages:

1. **adsorption** 吸附: reactant molecules stick to the catalyst surface.
2. the bonds in the reactants are weakened, so they react more easily.
3. **desorption** 脱附: the product molecules leave the surface.

Examples are iron in the Haber process, and platinum, palladium and rhodium in a catalytic converter.

Homogeneous catalysts

A **homogeneous catalyst** 均相催化剂 is in the same physical state as the reactants. It is used up in one step and then reformed in a later step, so it comes back unchanged. Examples are oxides of nitrogen helping to oxidise atmospheric sulfur dioxide, and Fe^{2+} or Fe^{3+} speeding up the reaction between I^- and $\text{S}_2\text{O}_8^{2-}$.